

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC304 / 20EC304 – NETWORK THEORY

17EC303 - NETWORK ANALYSIS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BTL* CO* PO*

1. a) Find an equivalent resistance between the terminals A and B as shown in the network Figure 1(a).

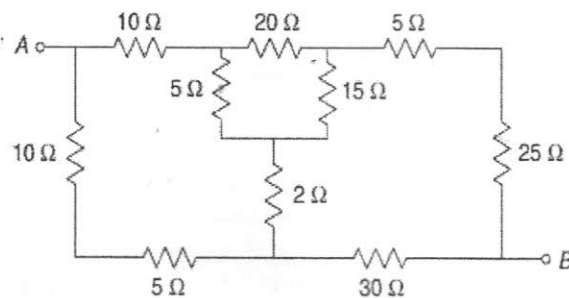


Figure 1.(a)

10 L*3 1 1

- b) For the network of Figure. 1.(b), find current through the 8Ω resistor using mesh analysis.

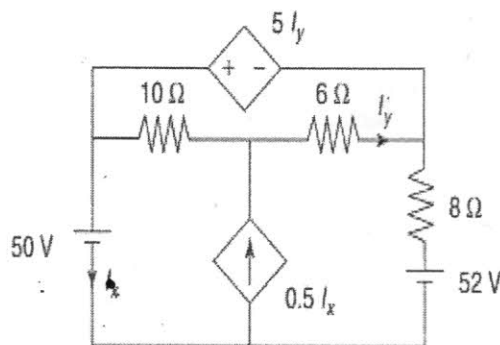


Figure 1.(b)

10 L3 1 1

2. a) Using mesh analysis find the current through the 5Ω resistor as shown in Figure. 2(a).

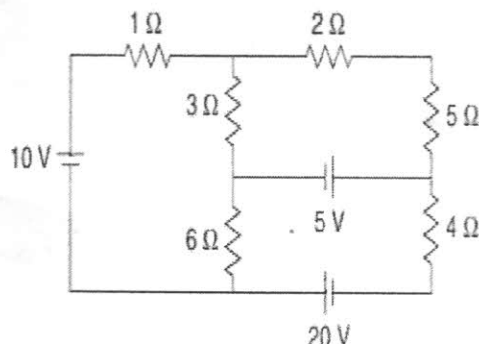


Figure 2.(a)

10 L3 1 1

21EC304/20EC304(17EC303)

Suppl Exam - Dec 2025

- b) For the network shown in Figure. 2.(b) find the node voltages V_1 and V_2 .

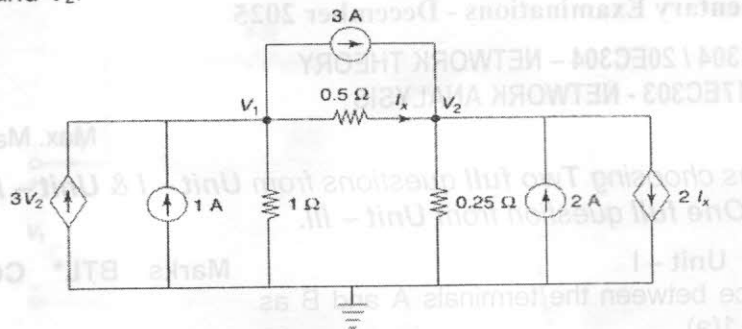


Figure 2.(b)

10 L3 1 1

3. a) For this network shown in Figure. 3.(a) and its associated oriented graph shown in Fig 3.(b) write down the tie-set matrix and obtain the network equilibrium equation in matrix form using KVL. Calculate the loop currents. Assume the Links as {1,2,3}.

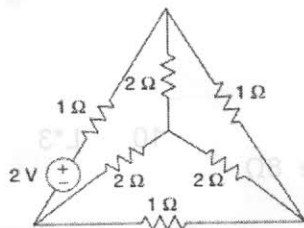


Figure 3.1(a)

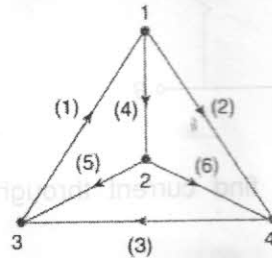


Figure 3.1(b)

10 L3 2 1

- b) For the network shown in Fig. 3.2(a) and its associated oriented graph shown in Fig 3.2(b), write down the cut-set matrix and obtain the network equilibrium equation in matrix form using KCL. Assume the Twigs as {2,4}.

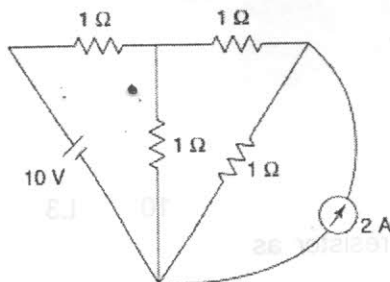


Figure 3.2(a)

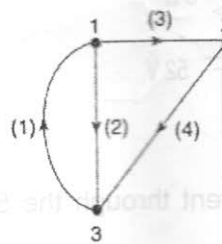


Figure 3.2(b)

10 L3 2 1

Unit - II

4. a) Using Superposition theorem, find the current through the 4 ohm resistor as shown in Figure. 4(a)

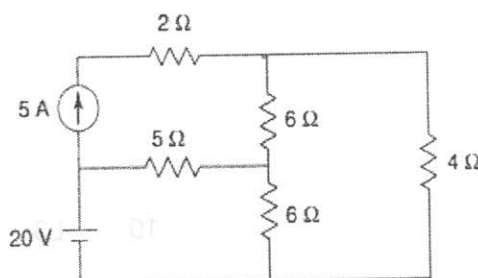


Figure 4.(a)

10 L3 3 1

- b) Find the current through the 2Ω resistor connected between terminals A and B as shown in the Figure. 4(b) using Thevenin's theorem.

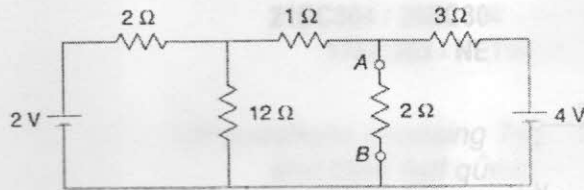


Figure 4.(b)

5. a) For the network shown in Figure. 5.(a), find the voltage V_x and verify the reciprocity theorem.

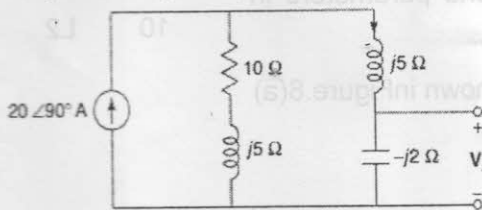


Figure. 5.(a)

- b) Find the current through the 10Ω resistor for the network shown in Figure. 5.(b), using Millman's theorem.

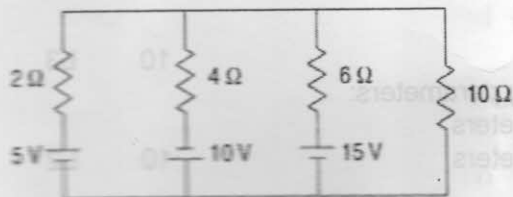


Figure 5.(b)

6. a) In the network shown in Figure. 6.(a), the switch is changed from the position 1 to the position 2 at $t = 0$, steady condition having reached before switching. Find the values of i , di/dt and d^2i/dt^2 at $t=0^+$.

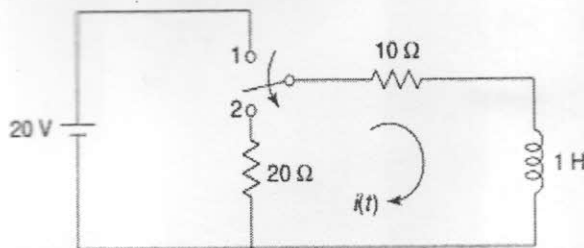


Figure 6(a)

- b) For the network shown in Figure. 6.(b), the switch is closed at $t = 0$, determine v , dv/dt and d^2v/dt^2 at $t=0^+$.

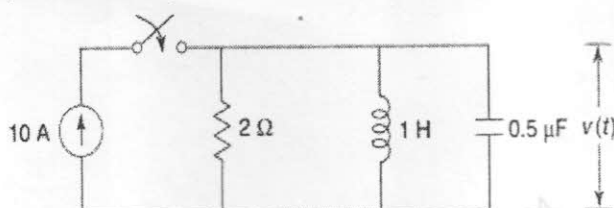


Figure. 6(b)

Unit – III

7. a) Find the Z- parameters for the network shown in Figure. 7(a)

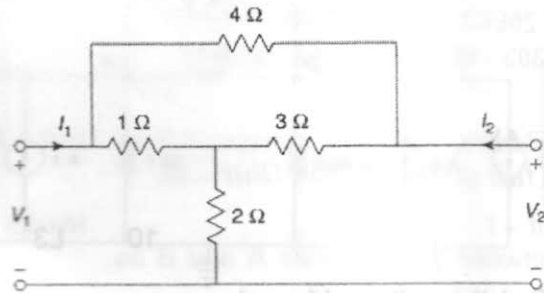


Figure. 7(a)

- b) Briefly explain hybrid parameters. Express hybrid parameters in terms of ABCD parameters.

8. a) Determine Z and Y-parameters of the network shown in Figure.8(a)

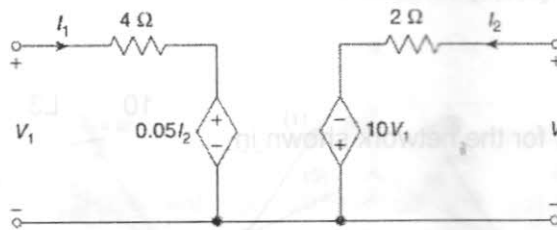


Figure 8.(a)

- b) Express the relationship between the following parameters:
- Y parameters in terms of hybrid parameters
 - ABCD parameters in terms of Z parameters

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
